EXPERIMENT 1

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1. Aim of the Experiment:

a. Implementation of Python Basic Libraries such as
   - Math
   - NumPy
   - Matplotlib
   - Seaborn
   - SciPy.

b. Creation and loading different Datasets in Python using JupyterNotebook.

c. Write a program to implement Linear regression on Jupyternotebook.
   - Download different datasets from Kaggle or UCI ML Repository.
   - Import all the necessary modules for the datasets.
   - Find out the best fit line along with the MSE, RMSE.

2. Steps Involved in Experiment/Practical:

a. Implementation of Python Basic Libraries such as (Math

```
import math
print("Square root of 16 using math module:", math.sqrt(16))
```

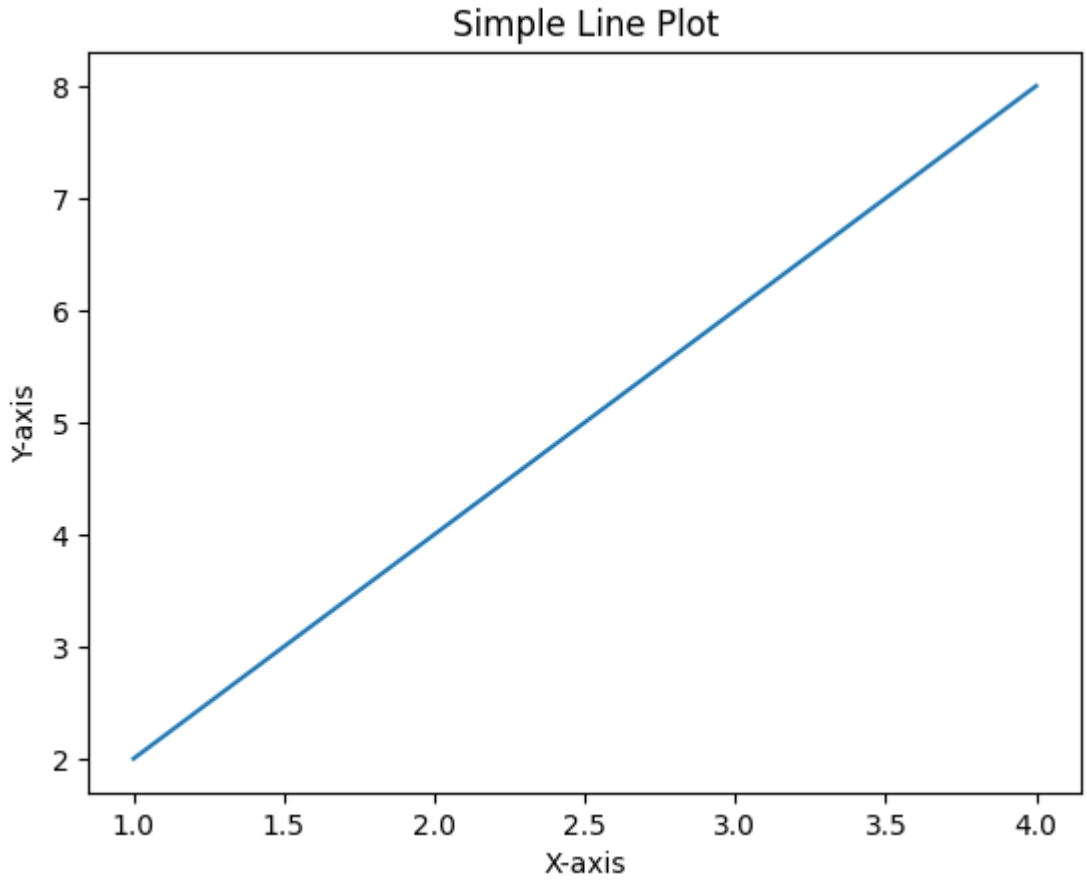
Output: Square root of 16 using math module: 4.0

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print("NumPy Array:", arr)
```

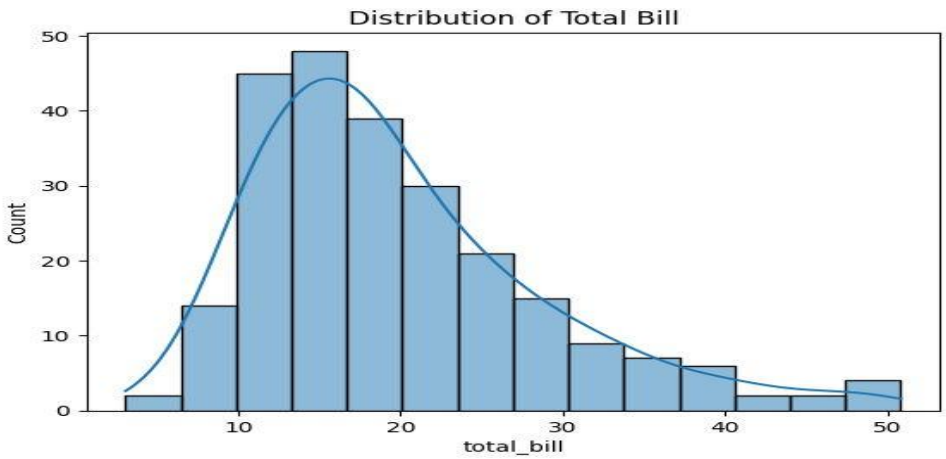
Output: NumPy Array: [1 2 3 4 5]

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4]
y = [2, 4, 6, 8]
plt.plot(x, y)
plt.title("Simple Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()
```

Output:



Output:



```
from scipy import stats
print("Mean using SciPy:", stats.tmean([1, 2, 3, 4, 5]))
```

Output: Mean using SciPy: 3.0

b. Creation and loading different Datasets in Python using JupyterNotebook

```
import pandas as pd
df = pd.read_csv('D:/3rd Sem/ML LAB/Code/annual-enterprise-survey- 2024-financial-year-provisional.csv')
df.head()
```

Output:

	Year	(All)
0	Variable_category	(All)
1	Units	Percentage
2	NaN	NaN
3	Row Labels	NaN
4	Accommodation	NaN

```
data = {
    'Name': ['Abhishek', 'Prachi', 'Gaurav','Om Priyanshu'], 'Age':
    [22,21,22,20],
    'City': ['Sasaram', 'Bettiah', 'Patna','Bokaro']
}
df=pd.DataFrame(data)
print(df)
```

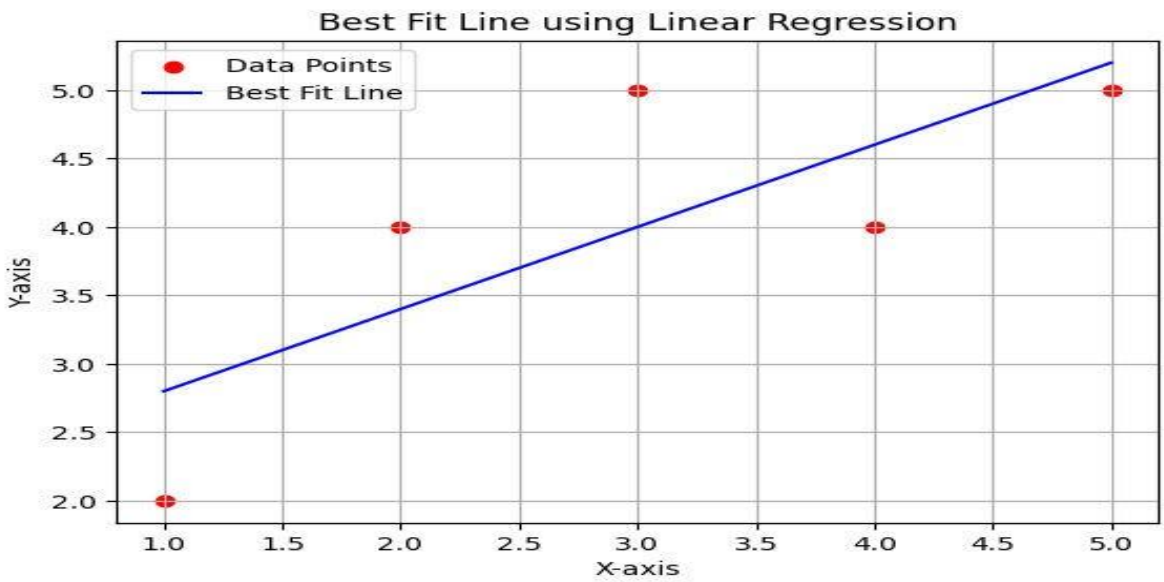
Output:

```
           Name  Age     City
0      Abhishek   22  Sasaram
1        Prachi   21  Bettiah
2        Gaurav   22    Patna
3  Om Priyanshu   20   Bokaro
```

c. Write a program to implement Linear regression on Jupyternotebook

```
# Step 5: Plot the original data and the best fit line
plt.scatter(X, Y, color='red', label='Data Points') points              # Original data

plt.plot(X, Y_pred, color='blue', label='Best Fit Line')  # Regression line
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Best Fit Line using Linear Regression")
plt.legend()
plt.grid(True)
plt.show()
```

Output:



```
mse = np.mean((Y - Y_pred) ** 2)
rmse = np.sqrt(mse)

print("Mean Squared Error (MSE):", mse) print("Root
Mean Squared Error (RMSE):", rmse)
```

Output:

Mean Squared Error (MSE): 0.47999999999999987

Root Mean Squared Error (RMSE): 0.6928203230275508

- Use the USA Housing Dataset (features: average income, number of rooms, bedrooms, population, house age, median house value).

```
 import pandas as pd
import numpy as np
 import matplotlib.pyplot as plt
 import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
 # Load dataset
df = pd.read_csv("data/USA_Housing.csv")  # make sure this file is in the working directory
df.head()
```

Output:

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price	Address
0	79545.458574	5.682861	7.009188	4.09	23086.800503	1.059034e+06	208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1	79248.642455	6.002900	6.730821	3.09	40173.072174	1.505891e+06	188 Johnson Views Suite 079\nLake Kathleen, CA...
2	61287.067179	5.865890	8.512727	5.13	36882.159400	1.058988e+06	9127 Elizabeth Stravenue\nDanieltown, WI 06482...
3	63345.240046	7.188236	5.586729	3.26	34310.242831	1.260617e+06	USS Barnett\nFPO AP 44820
4	59982.197226	5.040555	7.839388	4.23	26354.109472	6.309435e+05	USNS Raymond\nFPO AE 09386

#

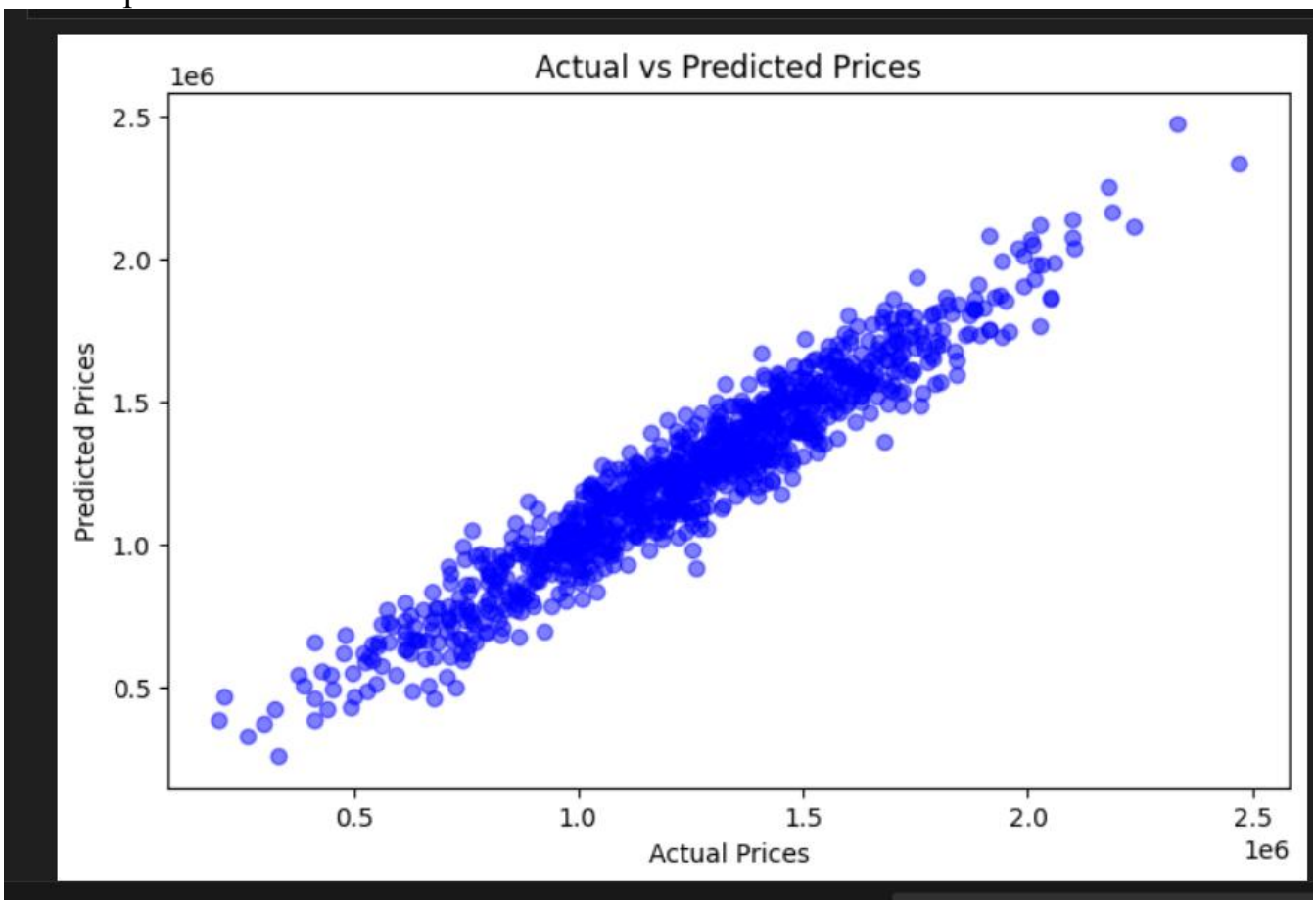
```
 # Drop 'Address'
df = df.drop('Address', axis=1)

# Features and Target
X = df.drop('Price', axis=1)
y = df['Price']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)

# Predicting on test data
y_pred = model.predict(X_test)
```

```
mse = mean_squared_error(y_test, y_pred)
print(f"Mean Squared Error (MSE): {mse}")
```

Output:

```
[7]
...  Mean Squared Error (MSE): 10089009300.894522
```

Output:



3. Learning Outcomes:

1. Understand how to perform data handling and preprocessing using NumPy for numerical computations.
2. Gain hands-on experience with data visualization techniques using Matplotlib.
3. Implement **CRUD operations** (Create, Read, Update, Delete) to manage real-world data structures effectively.
4. Implement linear regression from scratch using mathematical formulas.
5. Learn to evaluate model performance using error metrics such as MSE and RMSE.